# Semi-supervised Learning with Label Proportion

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**Abstract**—The scarcity of labels is common and great challenge in traditional supervised learning. Semi-supervised learning (SSL) leverages unlabeled samples to alleviate the absence of label information. Nevertheless, similar with annotation, label proportion is another type of prior information and plays a significant role in classification tasks. Compared with the acquisition of labels, label proportion can be obtained more easily. For example, only a small number of patients have been diagnosed with or not with cancers in hospital database. Meanwhile, the proportion with cancer can be generally estimated by historical records. How to incorporate such prior information of label proportion is crucial but rarely studied in literature. Traditional SSL methods often ignore this prior information and will lead to significant performance degradation. To solve this problem, we propose a novel SSL with Label Proportion. Our approach encourages to preserve label consistency and label proportion by imposing the cardinality bound constraints. Our formulated problem equals to a mixed-integer constrained submodular minimization and be difficult to solve directly. Therefore, we transformed the original problem into a convex one by Lovász extension and designed an efficient algorithm to solve it. Extensive experimental results present the improved performance of our method over several state-of-the-art methods.

**Index Terms**—Semi-supervised Learning, Label Proportion, Constrained Submodular Minimization, Imbalanced Classification.

## 1 Introduction

For the classification tasks, the features of each instance and its corresponding label are desired for supervised learning methods. These methods often try to learn the knowledge from data by characterizing the prior information, like the sparse and low-rank structure between features [1], [2], the distribution of instances [3] and the correlations of label [4]. However, due to the difficult acquisition of the labels, it is very common that there are only a few labeled samples and a large number of unlabeled instances in practice. This is a typical semi-supervised learning scenario and leads to the sharp degradation of supervised methods' performance due to lacking enough labels. Besides, in many real-world applications, similar with the annotation of instances and the structure of features, the label proportion of each class is another type of important prior and weak supervised information, which plays a significant role in classification tasks. For example, in inchoate diagnosis of various cancers, there are only a tiny part of patients diagnosed with cancer and the remaining large part of people are healthy or unknown cancer patients in the hospital database. Compared with the acquisition of accurate labels, the label proportion information is relatively easy collected. According to the GLOBOCAN 2018 estimates of cancer incidence and mortality produced by the International Agency for Research on Cancer [5], [6], there are 18,078,957 new cancer cases and 9,555,027 cancer deaths in 2018, which is equivalent to 238.4 and 125.8 per 100,000 population labeled with cancer, as shown in Table 1. In addition, Table 1 and Figure 1 present different type of cancers have different portions of incidence and mortality in different genders and regions. As known

Table 1
Incidence and mortality rates (per 100,000 population) of in total and gender in Northern America (NA), China and World Areas (WA) for All Cancers Combined in 2018, which is from the papers [5] and [6].

| Region | In Total | Gender | Incidence | Mortality |
|---|---|---|---|---|
| NA | 354.4/92.3 | Male | 387.6 | 104.2 |
| | | Female | 322.1 | 80.7 |
| China | 201.1/126.9 | Male | 223.0 | 166.6 |
| | | Female | 182.6 | 95.2 |
| WA | 238.4/125.8 | Male | 218.6 | 122.7 |
| | | Female | 182.6 | 83.1 |

that such label proportion information is very important to the diagnosis of cancers. To analyze this kind of data, there are at least two challenges: (1) lack of labeled data and (2) modeling the information of label portion. How to learn such prior information of label proportion is crucial but rarely studied. In this paper, we will proposed a novel method to solve these problems and improve the performance.

When we ignore the prior label proposition information, semi-supervised learning (SSL) [7], [8], [9] is an effective way to solve the problem with few labeled data and large unlabeled instances. By utilizing the features' structural information and a few of annotated labels, SSL is able to enhance the discriminative power of classifiers and has grown into one of important research area of machine learning and medical analysis. In literature, many semi-supervised methods are proposed. These existing semi-supervised classification algorithms can be divided into two categories according to their basic assumptions, i.e. the cluster assumption and the manifold assumption, which are consistent and belong to the generalization of smoothing assumption in supervised learning to semi-supervised learning [10]. The former assumes that the data has an inherent cluster structure, so instances that belong to the same cluster have the same class labels. The latter assumes that the data is on a manifold, so nearby instances have similar predictions.

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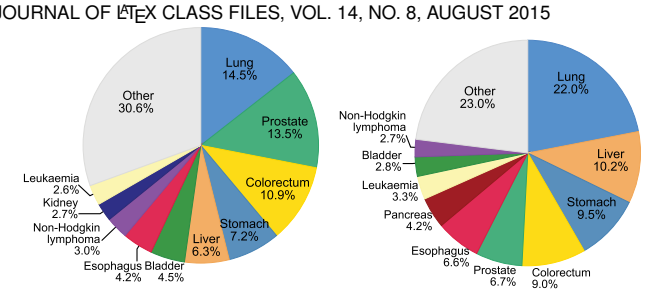
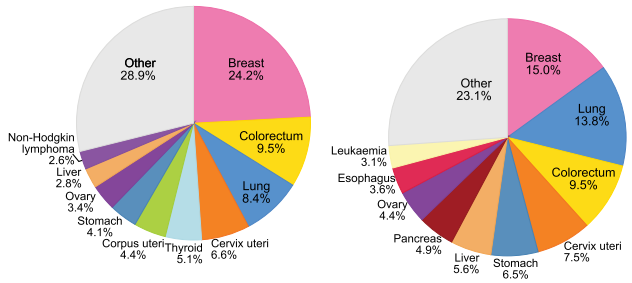



(a) Incidence (left) and Mortality (right) Distribution of Male




(b) Incidence (left) and Mortality (right) Distribution of Female

Figure 1. The distribution or portion of worldwide cases and deaths with the 10 most common cancers for male and female in 2018, which is from the papers [5] and [6].

Based on these two assumptions, there are four categories of semi-supervised learning approaches: generative methods, graph-based methods, disagreement-based methods and low-density separation methods.

For the generative method, labels of unlabeled samples can be estimated by EM (expectation maximization) algorithm or other approaches [11]. For example, Miller et al. utilized the maximization of total data likelihood to propose a classification model which is based on the joint probability model of features and labels [12]. Fujino et al. extended the generation mixture model by using the principle of maximum entropy through discriminant training, to combine the advantages of generative and discriminative methods [13]. There are many problems in this kind of algorithm, such as the local solution of EM algorithm. In general, domain knowledge is required to determine the appropriate generation model for good performance. Graph-based methods construct a graph, where the nodes correspond to training samples and the edges to the similarity or distance between instances, and then the label information is spread on graph according to certain criteria [14]. For example, labels can be propagated across different subgraphs separated by minimal cuts [15]. Apparently, how the graph is constructed will largely determine the performance of the algorithm. However, graph-based algorithms are transductive and computationally expensive so that they are difficult to adapt to new instances and big data. Disagreement-based methods assumes that there are multiple, different learners trained on the same labeled data [16]. The most famous representative is co-training [17], which is achieved by training two learners from two different feature sets. Based on canonical correlation analysis, Zhou et al. proposed a co-training algorithm requiring only one labeled point, which shows that co-training can be quite effective [18]. Low-density separation methods try to let the classification boundary through the lower density area [19]. The most famous example are S3VMs (semi-supervised support vector machines) [20], which aims to approach the optimal low-density separator. Since then, many scholars have made further studies, mainly focusing on solving S3VM non-convex problems [21], [22] and improving the robustness of the algorithms [23], [24], [25], [26].

Due to dealing with the scenario of few labeled and large amount of unlabeled instances effectively, SSL methods [8], [11], [21] pay more attention over the past few decades and have been successfully applied to many applications, such as image retrieval, text classification and machine translation. However, traditional SSL methods often ignore the prior information of label proportion. Due to ignoring the label proportion of each semantic class, these learning methods will not only obtain the biased models, which make the ratio of each class far away from the real label proportion during predicting testing data, but also lead to the sharp degradation of their generalization performance for classification.

To solve this problem, we propose a novel Semi-Supervised Learning method with Label Proportion (SSLLP), under general empirical risk minimization (ERM) framework in this paper. Specifically, an undirected graph in the sample space are constructed to pursuing the smoothness of the label in the semantic space. Besides, we propose to use the low-rank structure to depict label correlations and enhance the robustness to the noisy labels and outliers. Meanwhile, SSLLP attempts to encourage label consistency and label proportion information by imposing the cardinality bound constraints. The formulated problem is a mixed-integer constrained submodular minimization and difficult to solve directly. We further transform the original problem into a convex optimization problem by Lovász extension of submodular function and proposed to an efficient algorithm by incorporating conjugate gradient (CG) [27] and alternative direction method of multipliers (ADMM) [28]. We also provide the theoretical results, such as convergence analysis, computational complexity and parameter determination. Extensive experimental results present the improved performance of our SSLLP over several state-of-the-art methods. Our contributions are summarized as follows.

(1) We propose the SSLLP approach to solve this crucial, but rarely studied problem. As far as we know, this is the first research about the semi-supervised learning problem with the prior label proportion.
(2) We propose to tackle this problem by a mixed-integer constrained submodular minimization and transform it into a convex optimization problem by Lovász extension of submodular function.
(3) We develop an efficient algorithm to address the optimization problem and analyze its convergence behavior and computational complexity.
(4) We evaluate SSLLP systematically on eight real data sets from a broad range. Experimental results present that SSLLP outperforms other compared algorithms in almost all cases.

The rest of this paper is organized as follows. First, we introduce related work in Section 2. Section 3 will formulate this problem and we propose the SSLLP approach and some theoretical results to the transformation of our mixed-integer constrained submodular minimization by Lovász extension. The optimization of SSLLP is introduced in Section 4. And

then we provide some theoretical results about the parameter determination, convergence analysis and computational complexity. Experimental results on benchmark datasets are displayed to verify the effectiveness of SSLLP in Section 5. Finally, we conclude this paper in Section 6.

## 2 Related Work

In this section, we briefly review semi-supervised classification methods and the concept of submodular function in machine learning. Throughout this paper, all data points and the corresponding feature vectors are represented in the form of column vectors by boldface lowercase and matrices by boldface capital letters. Let $\mathbf{X} = (\mathbf{x}_1, \cdots, \mathbf{x}_l, \mathbf{x}_{l+1}, \cdots, \mathbf{x}_n)^\top$ be the samples feature and $\mathbf{Y} = (y_1, \cdots, y_l)^\top$ be the label matrix corresponding to the annotated samples, where $\mathbf{x}_i \in \mathbb{R}^d$ and $y_i \in \{0,1\}^m$ are the features and the label of the $i$-th sample, respectively. For any $j \in [m], y_{i,j} = 1$ if $\mathbf{x}_i$ is labeled as $j$ and $y_{i,j} = 0$ otherwise. Let $u = n - l$ be the number of unlabeled data points.

### 2.1 Semi-supervised Classification

For semi-supervised learning problem, one of the direct ways is to transform it into supervised learning case. Motivated by this idea, Chapelle et al. [19] proposed semi-supervised support vector machines (S3VMs) method under the low-density assumption. S3VMs selected the decision boundaries across low-density regions and found the optimal classification model by minimizing the following objective:

$$h(f,\hat{\mathbf{y}}) = \frac{\|f\|_{\mathcal{H}}}{2} + C_1 \sum_{i=1}^{l} \ell\left(y_i, f\left(\mathbf{x}_i\right)\right) + C_2 \sum_{j=1}^{u} \ell\left(\hat{y}_j, f\left(\hat{\mathbf{x}}_j\right)\right),$$

where $\mathcal{H}$ is the Reducing Kernel Hilbert Space (RKHS) and $\ell(\cdot)$ represents the hinge loss. To avoid trivial solution [29], S3VMs introduced a constraint $\hat{\mathbf{y}} \in \mathcal{B}$ and $\mathcal{B} = \{\hat{y} \in \{\pm 1\}^u | -\beta \leq \frac{\sum_{i=1}^{u} \hat{y}_j}{u} - \frac{\sum_{i=1}^{l} y_i}{u} \leq \beta\}$. Then, to take advantage of multiple candidate low-density separators and then select representative ones, Li et al. [21] proposed a safe semi-supervised support vector machine (S4VM). Hu et al. [22] combined S3VM with multiple kernel learning to achieve a combination of cluster assumption and manifold assumption. Recently, Zhang et al. [25] proposed semi-supervised optimal margin distribution machines that guaranteed optimization of the margin distribution rather than maximizing the minimum margin.

In addition, some other types of approaches are used to deal with semi-supervised problems. For example, a class of graph-based methods can be roughly addressed as the following optimization problems:

$$\min_{\hat{\mathbf{Y}} \in \mathbb{R}^{n \times m}} \operatorname{tr}\left(\hat{\mathbf{Y}}^T \mathbf{M} \hat{\mathbf{Y}} + \gamma (\hat{\mathbf{Y}} - \mathbf{Y})^T \mathbf{D} (\hat{\mathbf{Y}} - \mathbf{Y})\right)$$

where $\mathbf{M}$ can be thought of as a symmetric normalized Laplacian, $\mathbf{D}$ is a diagonal matrix whose diagonal elements are one for labeled data and zero for unlabeled data. A key task in this problem is to construct the matrix $\mathbf{M}$. To solve it, Zhou et al. [14] proposed local and global consistency (LGC) to construct the normalized Laplacian graph. Later, Xiang et al. [30] proposed semi-supervised local spline regression (SLSR) model to improve the quality of graph. The main idea of SLSR is to introduce splines in Sobolev space and map data points as class labels directly. Additionally, to improve the robustness of noisy case, Nie et al. [31] proposed general graph-based semi-supervised learning (GGSL) to learn the label information of unlabeled instances and find the potential new classes by random walks. Although these methods have applied into handwritten recognition and text classification, they are under the assumption of approximately equal portion proportion of each class and ignore label proportion information. However, there are many unbalanced cases with label proportion information, which cause a large deviation in the estimation of learning methods [19], [32] in real applications. Nevertheless, similar with the annotation of instances, the label proportion is another type of prior information and can play a significant role in many real classification tasks. To solve this problem, we propose a novel semi-supervised learning method to utilize the information of label proportion.

### 2.2 Submodular Function

In real applications, many learning problems can be formulated as the minimization problem of set functions, which is defined on the sunset of a given bases. Moreover, due to the relationship with discrete optimization, submodular functions play an important role in machine learning. In the following, we briefly introduce the concept of submodular functions and its applications.

**Define 1.** *A set function $F : 2^V \to \mathbb{R}$ is submodular if and only if, for all subsets $A, B \subseteq V$, we have:*

$$F(A) + F(B) \geqslant F(A \cup B) + F(A \cap B),$$

*where $2^V$ is the power set of base set $V$.*

In optimization theory we know that convex optimization is relatively easy to solve, actually there is a certain correlation between the submodular function and convex function. Through the Lovász extension, we can make the submodular function equivalent to the convex function.

**Define 2.** *Given a set function $F$ such that $F(\varnothing) = 0$, the Lovász extension $f : \mathbb{R}^p \to \mathbb{R}$ $(p = |V|)$ is defined as follows; for $w \in \mathbb{R}^p$ order the components in decreasing order $w_{j_1} \geqslant \cdots \geqslant w_{j_p}$, where $(j_1, \ldots, j_p)$ is a permutation, and define $f(w)$ through any of the following equivalent equations:*

$$f(w) = \sum_{k=1}^{p} w_{j_k} \left[F\left(\{j_1, \ldots, j_k\}\right) - F\left(\{j_1, \ldots, j_{k-1}\}\right)\right],$$

$$f(w) = \sum_{k=1}^{p-1} F\left(\{j_1, \ldots, j_k\}\right)\left(w_{j_k} - w_{j_{k+1}}\right) + F(V) w_{j_p},$$

$$f(w) = \int_{\min\{w_1, \ldots, w_p\}}^{+\infty} F(\{w \geqslant z\}) dz + F(V) \min\{w_1, \ldots, w_p\},$$

$$f(w) = \int_{0}^{+\infty} F(\{w \geqslant z\}) dz + \int_{-\infty}^{0} [F(\{w \geqslant z\}) - F(V)] dz.$$

In fact, there are many types of set functions that can be considered submodular, such as the cardinality-based functions, cut functions and set cover functions, which are widely used in machine learning. Some special cases are of interest, cardinality-based functions can be used for counting

problems in machine learning, some examples of restricted combinations of weighted cardinal functions can also be found [33]. Finding the minimum cut in undirected graph has become an important tool in machine learning, which is commonly called graph cut technique. Lovász extension in undirected graphs has become a classical regularization method [34]. Set cover functions are also often used as regular set-covers or sparsity-inducing norms [35].

## 3 Proposed Algorithm

We consider the generic empirical risk minimization (ERM) framework from the perspective of supervised learning to predict the labels of sample. Our prediction will be parameterized as $f(\mathbf{x};\mathbf{W}) = \mathbf{x}^\top\mathbf{W}$, where $\mathbf{W} \in \mathbb{R}^{d\times m}$. Although the number of instance features and labels may be large, there usually exist significant label correlations. This limits $\mathbf{W}$ to be a low-rank matrix, which not only controls overfitting, but also reduces the amount of computation, especially when it comes to matrix operations, like inverse matrices. Here we only consider the simplest linear model, but what we need to know is that our model can be easily generalized to the nonlinear case by simply applying a nonlinear transformation.

Naturally, we need to consider the consistency of the labels, that is, we want to minimize $\|\mathbf{Z}-\mathbf{X}\mathbf{W}\|_F^2$, Where $\mathbf{Z}$ is the label matrix including the labeled and unlabeled part. In addition, we adopt the information of label smoothness which is based on the assumption that if two instances are close enough to each other in the feature space, they should also have similar labels, i.e.,

$$\mathrm{tr}(\mathbf{Z}^\top\mathbf{L}\mathbf{Z}) = \frac{1}{2}\sum_{i,j=1}^{n}\mathbf{G}_{i,j}\|\mathbf{Z}_i-\mathbf{Z}_j\|_2^2$$

where $N_k(x_i)$ represents the set of the $k$ nearest neighbors of instance $x_i$, $\mathbf{G}$ is the instance similarity matrix with elements $\mathbf{G}(i,j) = \exp\left(-\frac{\|(\mathbf{x}_i-\mathbf{x}_j)\|^2}{\delta^2}\right)$ for $\mathbf{x}_j \in N_k(\mathbf{x}_i)$ or $\mathbf{x}_i \in N_k(\mathbf{x}_j)$ and the else are 0, $\delta$ is the kernel size. The normalized Laplacian matrix is $\mathbf{L} = \mathbf{D}-\mathbf{G}$ with $\mathbf{D} = \mathrm{diag}\,(d_1,\cdots,d_n)$, and $d_i = \sum_{j=1}^{n}\mathbf{G}_{ij}$. The weighted graph corresponding to the matrix $\mathbf{G}$ is $\mathcal{G} = (\mathcal{V},\mathcal{E})$ with $\mathcal{V} = \{1,\cdots,n\}$ and $\mathcal{E} = \{1,\cdots,n_e\}$. Where $n_e$ is the number of edges in the graph, which is equal to half the number of non-zero elements in $\mathbf{G}$. The edge between nodes $e_1$ and $e_2$ is $e \in \{1,\cdots,n_e\}$ with weight $G(e_1,e_2)$. When we minimize this term, it indicates that the higher the similarity of two instances are, the smaller the difference of their corresponding labels should be, which is consistent with our previous assumption.

At this point, we have the following model in the framework of ERM:

$$\min_{\mathbf{W},\mathbf{Z}} J(\mathbf{W},\mathbf{Z}) = \frac{1}{2}\|\mathbf{Z}-\mathbf{X}\mathbf{W}\|_F^2 + \lambda_1 r(\mathbf{W}) + \lambda_2\mathrm{tr}(\mathbf{Z}^\top\mathbf{L}\mathbf{Z}).$$

The second item $r(\mathbf{W}) : \mathbb{R}^{d\times L} \to \mathbb{R}$ is a regularizer, which limits the complexity of the model. Further, we consider the constraints of the model and additional information. For $\mathbf{W}$, we want it to have a small rank, namely $\mathrm{rank}(\mathbf{W}) \le k$, where $k \le \min\{m,n\}$.

For the label matrix $\mathbf{Z}$, we can make full use of the valid information that labels are proportional, i.e. $\mathbf{1}_n^\top\mathbf{Z} = \boldsymbol{v}^\top$, where $\boldsymbol{v}$ is a $m$-dimensional vector that limits the number of samples per class in the sample pool. Generally, we cannot know the exact number of samples in each category as it's hard to know in advance in practical applications. Fortunately, it is generally easy to know the approximate range of samples in each category, which is not accurate enough, but still useful in our model. This situation can also be divided into two cases, that is, the ratio of labels above or below a certain threshold, which can be formulated as $\mathbf{1}_n^\top\mathbf{Z} \ge \boldsymbol{v}_1^\top$ or $\mathbf{1}_n^\top\mathbf{Z} \le \boldsymbol{v}_2^\top$. For example, in spam analysis, we generally know that the amount of spam in all emails is less than a certain threshold, while in text analysis, we know that the frequency of some words is higher than a certain threshold. For all these cases, we can unify them as a bilateral constraint, that is our model include such information $\boldsymbol{v}_1^\top \le \mathbf{1}_n^\top\mathbf{Z} \le \boldsymbol{v}_2^\top$, where $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ are $m$-dimensional column vectors that limit the proportions of each labels.

Combined with the above information, we have:

$$\begin{aligned}&\min_{\mathbf{W},\mathbf{Z}} J(\mathbf{W},\mathbf{Z}) = \frac{1}{2}\|\mathbf{Z}-\mathbf{X}\mathbf{W}\|_F^2 + \lambda_1 r(\mathbf{W}) + \lambda_2\mathrm{tr}(\mathbf{Z}^\top\mathbf{L}\mathbf{Z}),\\ &\text{s.t. } \mathrm{rank}(\mathbf{W}) \le k, \mathbf{Z} \in \{0,1\}^{n\times m}, \boldsymbol{v}_1^\top \le \mathbf{1}_n^\top\mathbf{Z} \le \boldsymbol{v}_2^\top.\end{aligned} \tag{1}$$

So far, we can use alternating minimization technique to optimize $\mathbf{W}$ and $\mathbf{Z}$ respectively. Specifically, we need to optimize the following equations separately:

$$J_1(\mathbf{W}) = \frac{1}{2}\|\mathbf{Z}-\mathbf{X}\mathbf{W}\|_F^2 + \lambda_1 r(\mathbf{W}),\ \text{s.t. } \mathrm{rank}(\mathbf{W}) \le k. \tag{2}$$

$$\begin{aligned}&J_2(\mathbf{Z}) = \frac{1}{2}\|\mathbf{Z}-\mathbf{X}\mathbf{W}\|_F^2 + \lambda_2\mathrm{tr}(\mathbf{Z}^\top\mathbf{L}\mathbf{Z}),\\ &\text{s.t.}\mathbf{Z} \in \{0,1\}^{n\times m}, \boldsymbol{v}_1^\top \le \mathbf{1}_n^\top\mathbf{Z} \le \boldsymbol{v}_2^\top.\end{aligned} \tag{3}$$

For the label matrix $\mathbf{Z}$, there are two potential constraints that need our attention: 1) although the label matrix is constantly changing during the updating process, the part corresponding to the part that already has labels is fixed, i.e. $\mathbf{Z}_{\mathcal{L}} = \mathbf{Y}$, where $\mathcal{L}$ is the index set of labeled data. 2) since each sample only belongs to one category, the sum of label values of each sample is one. Taking these two things into account, we can obtain the new $J_2$ as follows:

$$\begin{aligned}J_2(\mathbf{Z}_{\mathcal{U}}) = &\frac{1}{2}\|\mathbf{Z}_{\mathcal{U}}-\mathbf{X}_{\mathcal{U}}\mathbf{W}\|_F^2 + \frac{\lambda_2}{2}(\sum_{i,j\in\mathcal{U}}\mathbf{G}_{i,j}\|\mathbf{Z}_i-\mathbf{Z}_j\|_2^2\\ &+ 2\sum_{i\in\mathcal{L},j\in\mathcal{U}}\mathbf{G}_{i,j}\|\mathbf{Z}_i-\mathbf{Z}_j\|_2^2) + C\end{aligned} \tag{4}$$

where $\mathcal{U}$ is the index set of unlabeled data, $C = \frac{1}{2}\|\mathbf{Z}_{\mathcal{L}}-\mathbf{X}_{\mathcal{L}}\mathbf{W}\|_F^2 + \frac{\lambda_2}{2}\sum_{i,j\in\mathcal{L}}\mathbf{G}_{i,j}\|\mathbf{Z}_i-\mathbf{Z}_j\|_2^2$ is a constant independent of $\mathbf{Z}_{\mathcal{U}}$. Further, since each row of $\mathbf{Z}$ has and only one element is 1 and the rest are 0, there is $\|\mathbf{Z}_i\|_2^2 = 1$ for all $i \in \mathcal{U}$, so

$$\begin{aligned}J_2(\mathbf{Z}_{\mathcal{U}}) = &-\mathrm{tr}[(\mathbf{X}_{\mathcal{U}}\mathbf{W})^\top\mathbf{Z}] + \frac{\lambda_2}{2}(\sum_{i,j\in\mathcal{U}}\mathbf{G}_{i,j}\|\mathbf{Z}_i-\mathbf{Z}_j\|_2^2\\ &- 4\mathrm{tr}[\mathbf{Y}^\top\mathbf{G}_{\mathcal{L}\mathcal{U}}\mathbf{Z}_{\mathcal{U}}]) + C'.\end{aligned} \tag{5}$$

$\mathbf{G}_{\mathcal{L}\mathcal{U}}$ is a subblock in the partitioned matrix $\mathbf{G}$ with elements $\mathbf{G}_{ij}$ where $i \in \mathcal{L}$ and $j \in \mathcal{U}$. Considering the constraints in

formula in Eq. (1), we get the final optimization problem for $\mathbf{Z}_{\mathcal{U}}$ as follows:

$$\min_{\mathbf{Z}_{\mathcal{U}}} F(\mathbf{Z}_{\mathcal{U}}) = \frac{\lambda_2}{2}\Big(\sum_{i,j\in\mathcal{U}} \mathbf{G}_{i,j}\left\|\mathbf{Z}_i - \mathbf{Z}_j\right\|_2^2 - \mathrm{tr}[(\mathbf{X}_{\mathcal{U}}\mathbf{W} + 2\lambda_2\mathbf{G}_{\mathcal{U}\mathcal{L}}\mathbf{Y})^\top\mathbf{Z}_{\mathcal{U}}], \text{ s.t. } \mathbf{Z}_{\mathcal{U}} \in \Omega_1. \tag{6}$$

$\Omega_1 = \{0,1\}^{u\times m} \cap \Gamma$, and $\Gamma = \{\mathbf{Z} \mid \boldsymbol{v}_l^\top - \mathbf{1}_l^\top\mathbf{Y} \le \mathbf{1}_u^\top\mathbf{Z} \le \boldsymbol{v}_u^\top - \mathbf{1}_l^\top\mathbf{Y}, \mathbf{Z}\cdot\mathbf{1}_m = \mathbf{1}_u\}$ denotes the row and column constraint of the label matrix for unlabeled data.

Problem in Eq. (6) is NP hard, which is difficult to solve directly. We will discuss three different approximate methods to solve it. The first is simply to remove the cardinality constraint, thus becoming a submodular optimization problem $\min_{\mathbf{Z}\in\{0,1\}^{u\times m}} F(\mathbf{Z})$, which can be precisely solved using the st-cut algorithm [36]. The second is a linear programming (LP) relaxation approximation of the first model based on the Lovász extension of submodular functions $\min_{\mathbf{Z}\in[0,1]^{u\times m}} f(\mathbf{Z})$(see Proposition 2). The two methods have a common shortcoming, that is, they can't deal with the linear constraint on the target variable reasonably. The third method, which is used in this paper, is to add cardinality constraint on the second method, i.e:

$$\min_{\mathbf{Z}} f(\mathbf{Z}_{\mathcal{U}}), \text{ s.t. } \mathbf{Z}_{\mathcal{U}} \in \Omega_2\,. \tag{7}$$

where $\Omega_2 = [0,1]^{u\times m} \cap \Gamma$.

**Proposition 1.** *Problem in Eq. (6) is a constrained submodular minimization (CSM) problem, and it is NP-hard.*

**Proposition 2.** *The Lovász extension of objective function in Eq. (6) is formulated as:*

$$f(\mathbf{Z}_{\mathcal{U}}) = -\mathrm{tr}[(\mathbf{X}_{\mathcal{U}}\mathbf{W} + 2\lambda_2\mathbf{G}_{\mathcal{U}\mathcal{L}}\mathbf{Y})^\top\mathbf{Z}_{\mathcal{U}}] + \lambda_2\sum_{i,j\in\mathcal{U}}\sum_{k}^{m}\mathbf{G}(i,j)|\mathbf{Z}_{ik} - \mathbf{Z}_{jk}| + C' \tag{8}$$

Before proving the propositions, we introduce some notations and lemmas. Let's denote $\mathbf{X}_{\mathcal{U}}\mathbf{W} + 2\lambda_2\mathbf{G}_{\mathcal{U}\mathcal{L}}\mathbf{Y}$ as $\mathbf{Y}_{\mathbf{w}}$ for simplicity, and we do the following transformation:

$$\begin{aligned}
&\boldsymbol{z} = vec(\mathbf{Z}_{\mathcal{U}}) = [\mathbf{Z}_{u+11},\cdots,\mathbf{Z}_{n1},\cdots,\mathbf{Z}_{nm}]^\top \in \{0,1\}^{mu},\\
&\boldsymbol{y}_{\mathbf{w}} = vec(\mathbf{Y}_{\mathbf{w}}) = [\mathbf{Y}_{\mathbf{w}11},\cdots,\mathbf{Y}_{\mathbf{w}u1},\cdots,\mathbf{Y}_{\mathbf{w}um}]^\top \in \mathbb{R}^{mu},\\
&\hat{\mathbf{G}} = \mathbf{I}_m \otimes \mathbf{G}_{\mathcal{U}\mathcal{U}} \in \mathbb{R}^{mu\times mu},\\
&\mathbf{H}_1 = \left[\mathbf{1}_u^\top, \mathbf{0}_{(m-1)u}^\top; \mathbf{0}_u^\top, \mathbf{1}_u^\top, \mathbf{0}_{(m-2)u}^\top; \ldots; \mathbf{0}_{(m-1)u}^\top, \mathbf{1}_u^\top\right]\\
&\quad \in \{0,1\}^{u\times mu},\\
&\mathbf{H}_2 = [\mathbf{I}_u,\ldots,\mathbf{I}_u] \in \{0,1\}^{u\times um}.
\end{aligned}$$

Where $\otimes$ denotes the Kronecker product. Note that $\hat{\mathbf{G}}_{\mathcal{U}\mathcal{U}}$ can be understood as the similarity matrix of a large graph with $mu$ nodes and $mu_e$ edges. And let $\hat{\mathbf{L}}$ be the corresponding Laplace matrix. Then, the problem in Eq. (6) can be reformulated as follows:

$$\begin{aligned}
&\min_{\boldsymbol{z}\in\{0,1\}^{um}} -\sum_{i}^{mu}\boldsymbol{z}_i\boldsymbol{y}_{\mathbf{w}_i} + \frac{\lambda_2}{2}\sum_{i,j}^{mu}\hat{\mathbf{G}}(i,j)\left[\boldsymbol{z}_i - \boldsymbol{z}_j\right]^2\\
&\equiv \min_{\boldsymbol{z}\in\{0,1\}^{mu}} F(\boldsymbol{z}) = -\boldsymbol{y}_{\mathbf{w}}^\top\boldsymbol{z} + \lambda_2\boldsymbol{z}^\top\hat{\mathbf{L}}\boldsymbol{z}\\
&\text{s.t. } \boldsymbol{v}_1 - \mathbf{Y}^\top\mathbf{1}_l \le \mathbf{H}_1\boldsymbol{z} \le \boldsymbol{v}_2 - \mathbf{Y}^\top\mathbf{1}_l, \mathbf{H}_2\boldsymbol{z} = \mathbf{1}_u.
\end{aligned} \tag{9}$$

**Lemma 1.** *Let $Q \in \mathbb{R}^{p\times p}$ and $q \in \mathbb{R}^p$, then the quadratic set function $F(A) = q^\top\mathbf{1}_A + \frac{1}{2}\mathbf{1}_A^\top Q\mathbf{1}_A$ is submodular if and only if the off-diagonal entries of $Q$ are non-positive. $\mathbf{1}_A \in \{0,1\}^p$ denotes the indicator vector of the subset $A$: if $i \in A$ then $\mathbf{1}_A(i) = 1$, otherwise, $\mathbf{1}_A(i) = 0$. (See Proposition 6.3 in [37]).*

*Proof of Proposition 1.* Firstly, define a set function $F : V \to \mathbb{R}$ with $V = \{1,\cdots,mu\}$. Given a subset $A \subset V$, and $\mathbf{z}$ can be seen as the sign vector of $A$, when $i \in A$, then $\boldsymbol{z}(i) = 1$, otherwise $\boldsymbol{z}(i) = 0$. Substitute it into Eq. (9), we obtain

$$-\boldsymbol{y}_{\mathbf{w}}^\top\boldsymbol{z} + \lambda_2\mathbf{z}^\top\hat{\mathbf{L}}\mathbf{z} = -\boldsymbol{y}_{\mathbf{w}}^\top\mathbf{1}_A + \lambda_2\mathbf{1}_A^\top\hat{\mathbf{L}}\mathbf{1}_A = F(A)$$

Obviously, $\hat{\mathbf{L}}$ is a normalized graph Laplacian matrix so that its off-diagonal entries are non-positive. As we know from lemma 1, $F$ is equivalent to a submodular function. The constraint $\boldsymbol{v}_1 - \mathbf{Y}^\top\mathbf{1}_l \le \mathbf{H}_1\boldsymbol{z} \le \boldsymbol{v}_2 - \mathbf{Y}^\top\mathbf{1}_l, \mathbf{H}_2\boldsymbol{z} = \mathbf{1}_u$ can be seen as the cardinality bounds of $\mathbf{z}$. In addition, Eq. (9) is also a cut function, if the objective function is a cut and submodular function, then its minimization with cardinality constraint is NP-hard [37], [38]. Therefore, we conclude that the problem in Eq. (9) is NP-hard. Since this is equivalent to the problem in Eq. (6), it is also a NP-hard problem. □

*Proof of Proposition 2.* Firstly, we prove that the Lovász extension of the quadratic term $\mathbf{z}^\top\hat{\mathbf{L}}\mathbf{z}$ in Eq. (9) is:

$$\hat{f}(\boldsymbol{z}) = \sum_{i,j}^{mu}\hat{\mathbf{G}}(i,j)|\boldsymbol{z}_i - \boldsymbol{z}_j|.$$

Define a set function $\hat{F} : 2^V \to \mathbb{R}$ corresponding to the quadratic term in Eq. (9),

$$\begin{aligned}
\hat{F}(A) &= \frac{1}{2}\sum_{i,j}^{mu}\hat{\mathbf{G}}(i,j)\left[\boldsymbol{z}_i - \boldsymbol{z}_j\right]^2 = \frac{1}{2}\sum_{i,j\in A}\hat{\mathbf{G}}(i,j)\left[\boldsymbol{z}_i - \boldsymbol{z}_j\right]^2\\
&+ \frac{1}{2}\sum_{i,j\in V\setminus A}\hat{\mathbf{G}}(i,j)\left[\boldsymbol{z}_i - \boldsymbol{z}_j\right]^2 + \sum_{i\in A, j\in V\setminus A}\hat{\mathbf{G}}(i,j)\left[\boldsymbol{z}_i - \boldsymbol{z}_j\right]^2
\end{aligned}$$

For the first two terms, since $\boldsymbol{z}_i = \boldsymbol{z}_j$, we have $\hat{F}(A) = \sum_{i\in A, j\in V\setminus A}\hat{\mathbf{G}}(i,j)\left[\boldsymbol{z}_i - \boldsymbol{z}_j\right]^2$, which is a standard cut function. Thus the Lovász extension of $\hat{F}$ is $\hat{f}(\boldsymbol{z})$. Besides, the Lovász extension of the linear term $\boldsymbol{y}_{\mathbf{w}}^\top\boldsymbol{z}$ still has the same form. Thus the Lovász extension of the objective function in Eq. (6) is

$$f(\boldsymbol{z}) = -\boldsymbol{y}_{\mathbf{w}}^\top\boldsymbol{z} + \lambda_2\hat{f} = -\boldsymbol{y}_{\mathbf{w}}^\top\boldsymbol{z} + \lambda_2\sum_{i,j}^{mu}\hat{\mathbf{G}}(i,j)|\boldsymbol{z}_i - \boldsymbol{z}_j|$$

Finally, it is easy to get Eq. (8) by using the transformation between variables. □

It is noted that in this model, we take into account many necessary factors in semi-supervised learning, including some potential factors, and the label proportion constraint is a natural introduction in our model. From the generalization framework of the model, our model can easily be extended to nonlinear cases and other loss functions. In addition, compared with many traditional semi-supervised learning, we can not only realize label propagation more accurately, but also construct a simple and effective classifier, which makes our model very convenient to deal with the classification of new samples.

There are several key points to note in our model. Firstly, how shall we implement the label propagation and training of classifier at the same time, and secondly, how do we deal with low-rank constraints and discretization attributes of labels for these two processes. These are all important considerations in the solving process of the model. Aiming at these two points, in the section 4, the alternating iterative method based on low-rank decomposition and the ADMM method based on Lovász extension are respectively used to solve these two variables.

## 4 Optimization

In this section, we begin to solve the proposed model precisely. Because of the proportionality constraint and discretization property of $\mathbf{Z}$ as well as the low-rank constraint of $\mathbf{W}$, it is difficult to solve Eq. (1) directly. Therefore, we use the alternating iteration method to obtain their optimal.

### 4.1 Alternating Iteration Solution to W

Let's first think about how do we solve for the classifier $\mathbf{W}$. We consider its low-rank decomposition $\mathbf{W} = \mathbf{U}\mathbf{V}^\top$,where $\mathbf{U} \in \mathbb{R}^{d\times k}$, $\mathbf{V} \in \mathbb{R}^{m\times k}$. Let the regularizer $r(\cdot)$ be trace norm regularization, which can be decomposed as $\|W\|_{\text{tr}} = \frac{1}{2}\left(\|U\|_F^2 + \|V\|_F^2\right)$. Then, when $\mathbf{Z}$ is fixed, Eq. (1) is equivalent to minimizing over $\mathbf{U}$ and $\mathbf{V}$:

$$J_1(\mathbf{U},\mathbf{V}) = \frac{1}{2}\|\mathbf{Z} - \mathbf{X}\mathbf{U}\mathbf{V}^\top\|_F^2 + \frac{\lambda_1}{2}(\|\mathbf{U}\|_F^2 + \|\mathbf{V}\|_F^2) \quad (10)$$

Note that when either $\mathbf{U}$ or $\mathbf{V}$ is fixed, the optimal object $J_1(\mathbf{U},\mathbf{V})$ becomes a convex function. This allows us to apply alternating minimization to Eq. (10). After proper initialization, a sequence $\{(\mathbf{U}^t,\mathbf{V}^t)\}$ is generated by

$$\begin{aligned} \mathbf{V}^{(t)} &\leftarrow \arg\min_{\mathbf{V}} J_1(\mathbf{U}^{t-1},\mathbf{V}), \\ \mathbf{U}^{(t)} &\leftarrow \arg\min_{\mathbf{U}} J_1(\mathbf{U},\mathbf{V}^t). \end{aligned} \quad (11)$$

Once $\mathbf{U}$ is fixed, updating $\mathbf{V}$ is easy as $\mathbf{V}$ can be updated as follows:

$$\mathbf{V} = (\mathbf{Z}^\top\mathbf{X}\mathbf{U})[\lambda_1 I_k + (\mathbf{X}\mathbf{U})^\top\mathbf{X}\mathbf{U}]^{-1} \quad (12)$$

which is easy to solve as $k$ is small in general. While if $\mathbf{V}$ is fixed, updating $\mathbf{U}$ is more complicated, because all the variables are confused by multiplication with $\mathbf{X}$. Here we do the following transformation of the original problem. Let $\boldsymbol{u} = vec(\mathbf{U}), \boldsymbol{z} = vec(\mathbf{Z})$, Then $\boldsymbol{u}^* = \min_{\boldsymbol{u}\in\mathbb{R}^{dk}} \; g(\boldsymbol{u})$.

$$g(\boldsymbol{u}) \equiv \frac{1}{2}\|\boldsymbol{z} - (\mathbf{V}\otimes\mathbf{X})\boldsymbol{u}\|_2^2 + \frac{\lambda_1}{2}\|\boldsymbol{u}\|_2^2 \quad (13)$$

where $\otimes$ denotes the outer product. Similarly with the equation in Eq. (12), we have

$$\boldsymbol{u} = \left[\lambda_1 I_{kd} + (\mathbf{V}\otimes\mathbf{X})^\top(\mathbf{V}\otimes\mathbf{X})\right]^{-1}\cdot[(\mathbf{V}\otimes\mathbf{X})^\top\boldsymbol{z}] \quad (14)$$

However, it is inefficient to compute the closed form solution for the above problem. When $d$ is large, the above closed form solution requires inverting a $dk \times dk$ matrix which is hardly regarded as feasible. Instead, it is more appropriate to employ efficient gradient descent methods, e.g., conjugate gradient (CG), for optimization.

**Algorithm 1** Conjugate gradient for solving the problem (13)

**Set** initial $\boldsymbol{u}_0, \boldsymbol{r}_0 = -\nabla g(\boldsymbol{u}_0), \boldsymbol{d}_0 = \boldsymbol{r}_0$
**For** $t = 0, 1, 2, \cdots$
If $\|\boldsymbol{r}_t\|$ is small enough, then stop the procedure and return $\boldsymbol{u}_t$
$\alpha_t = \frac{\boldsymbol{r}_t^\top\boldsymbol{r}_t}{\boldsymbol{d}_t^\top\nabla^2 g(\boldsymbol{u})\boldsymbol{d}_t}$
$\boldsymbol{u}_{t+1} = \boldsymbol{u}_t + \alpha_t\boldsymbol{d}_t$
$\boldsymbol{r}_{t+1} = \boldsymbol{r}_t - \alpha_t\nabla^2 g(\boldsymbol{u})\boldsymbol{d}_t$
$\beta_t = \frac{\boldsymbol{r}_{t+1}^\top\boldsymbol{r}_{t+1}}{\boldsymbol{r}_t^\top\boldsymbol{r}_t}$
$\boldsymbol{d}_{t+1} = \boldsymbol{r}_{t+1} + \beta_t\boldsymbol{d}_t$

Obviously $g$ is a quadratic differentiable function, the gradient and the Hessian matrix for $g(\boldsymbol{u})$ are:

$$\begin{aligned} \nabla g(\boldsymbol{u}) &= [(\mathbf{V}\otimes\mathbf{X})^\top(\mathbf{V}\otimes\mathbf{X}) + \lambda_1 I_{kd}]\boldsymbol{u} - (\mathbf{V}\otimes\mathbf{X})^\top\boldsymbol{z}, \\ \nabla^2 g(\boldsymbol{u}) &= (\mathbf{V}\otimes\mathbf{X})^\top(\mathbf{V}\otimes\mathbf{X}) + \lambda_1 I_{kd}. \end{aligned} \quad (15)$$

A computation of $\nabla g(\boldsymbol{u})$ and $\nabla^2 g(\boldsymbol{u})$ directly using Eq. (15) takes a lot of time. Here we give faster procedures for performing these two operations.

Using the identity $\left(\mathbf{B}^\top\otimes\mathbf{A}\right)vec(\mathbf{X}) = vec(\mathbf{A}\mathbf{X}\mathbf{B})$, we have

$$\begin{aligned} \nabla g(\boldsymbol{u}) &= vec\left[\mathbf{X}^\top(\mathbf{X}\mathbf{U}\mathbf{V}^\top - \mathbf{Z})\mathbf{V} + \lambda_1\mathbf{U}\right], \\ \nabla^2 g(\boldsymbol{u})\boldsymbol{s} &= vec\left[\mathbf{X}^\top\mathbf{X}\mathbf{S}\mathbf{V}^\top\mathbf{V} + \lambda_1\mathbf{S}\right]. \end{aligned} \quad (16)$$

where $\mathbf{S}$ be the $d\times k$ matrix such that $\boldsymbol{s} = vec(\mathbf{S})$. So far, we have solved the problem in Eq. (10) completely by the Algorithm 1.

### 4.2 ADMM Solution to Z

As it mentioned in section 3, we don't need to learn the full label matrix, but only those of unlabeled data. Once $\mathbf{W}$ is fixed, we can get $\mathbf{Z}_{\mathcal{U}}$ by solving the following problem:

$$\mathbf{Z}_{\mathcal{U}}^* = \arg\min_{\mathbf{Z}_{\mathcal{U}}} f(\mathbf{Z}_{\mathcal{U}}), \text{ s.t. } \mathbf{Z}_{\mathcal{U}} \in \Omega_2\,.$$

In the following statement, for convenience, let's just substitute $\mathbf{Z}$ for $\mathbf{Z}_{\mathcal{U}}$ without disagreement. Besides, let denote $u_e$ as the number of edges that belong to $\mathcal{U}$ for all endpoints.

Notice that through the Lovász extension, $\mathbf{Z}$ is a continuous variable, which makes it very convenient for us to deal with. At the same time, the following proposition shows that the optimal values of the functions before and after the extension are mutually constrained, which means that it makes sense for us to do the extension.

**Proposition 3.** *$F(\mathbf{Z})$ and $f(\mathbf{Z})$ satisfy the conditions:*

$$\begin{aligned} \min_{\mathbf{Z}\in\Omega_1} f(\mathbf{Z}) &\geq \min_{\mathbf{Z}\in\Omega_1} F(\mathbf{Z}) \geq \min_{\mathbf{Z}\in[0,1]^{u\times m}} f(\mathbf{Z}) = \min_{\mathbf{Z}\in\{0,1\}^{u\times m}} F(\mathbf{Z}), \\ \min_{\mathbf{Z}\in\Omega_1} f(\mathbf{Z}) &\geq \min_{\mathbf{Z}\in\Omega_2} f(\mathbf{Z}) \geq \min_{\mathbf{Z}\in[0,1]^{u\times m}} f(\mathbf{Z}) = \min_{\mathbf{Z}\in\{0,1\}^{u\times m}} F(\mathbf{Z}). \end{aligned}$$

Before we prove it, we give the Lemma 2.

**Lemma 2.** *Let $F : 2^V \to \mathbb{R}$ be a submodular function with $V = \{1, \cdots, p\}$, and $f$ its Lovász extension; then we have*

$$\min_{A\subset V} F(A) = \min_{\mathbf{a}\in\{0,1\}^p} F(\mathbf{a}) = \min_{a\in[0,1]^p} f(\mathbf{a})$$

*(See Proposition 3.7 in [37] ).*

*Proof of Proposition 3.* According to the above transformation and Lemma 2, it is easy to obtain

$$\min_{\mathbf{Z}\in[0,1]^{u\times m}} f(\mathbf{Z}) = \min_{\mathbf{Z}\in\{0,1\}^{u\times m}} F(\mathbf{Z}).$$

As $\Omega_1 \subset \{0,1\}^{u\times m}$, we have

$$\min_{\mathbf{Z}\in\Omega_1} F(\mathbf{Z}) \geq \min_{\mathbf{Z}\in\{0,1\}^{u\times m}} F(\mathbf{Z}) = \min_{\mathbf{Z}\in[0,1]^{u\times m}} f(\mathbf{Z}).$$

As $\Omega_2 \subset [0,1]^{u\times m}$, we have

$$\min_{\mathbf{Z}\in\Omega_2} f(\mathbf{Z}) \geq \min_{\mathbf{Z}\in[0,1]^{u\times m}} f(\mathbf{Z}) = \min_{\mathbf{Z}\in\{0,1\}^{u\times m}} F(\mathbf{Z}).$$

As $\Omega_1 \subset \Omega_1$, we have

$$\min_{\mathbf{Z}\in\Omega_1} f(\mathbf{Z}) \geq \min_{\mathbf{Z}\in\Omega_2} f(\mathbf{Z}).$$

We just have to prove

$$\min_{\mathbf{Z}\in\Omega_1} f(\mathbf{Z}) \geq \min_{\mathbf{Z}\in\Omega_1} F(\mathbf{Z})$$

We prove the following inequality $f(\mathbf{Z}) = f(\boldsymbol{z}) \geq F(\boldsymbol{z}) = F(\mathbf{Z})$ holds for all $\boldsymbol{z} \in [0,1]$, i.e.,

$$\begin{aligned} f(\boldsymbol{z}) - F(\boldsymbol{z}) &= \lambda_2(\hat{f}(\boldsymbol{z}) - \boldsymbol{z}^\top \hat{\mathbf{L}}\boldsymbol{z}) \\ &= \frac{\lambda_2}{2}\sum_{i,j=1}^{mu} \hat{\mathbf{G}}_{i,j}\left[2|\boldsymbol{z}_i - \boldsymbol{z}_i| - (\boldsymbol{z}_i - \boldsymbol{z}_i)^2\right] \\ &= \frac{\lambda_2}{2}\sum_{i,j=1}^{mu} \hat{\mathbf{G}}_{i,j}\left[|\boldsymbol{z}_i - \boldsymbol{z}_i|(2 - |\boldsymbol{z}_i - \boldsymbol{z}_i|)\right] \geq 0 \end{aligned}$$

Furthermore, as $\Omega_1 \subset \{0,1\}^{u\times m}$, the last inequality is proved. Finally all proofs are finished. □

As demonstrated in [37], a set-function $F$ is submodular if and only if its Lovász extension $f$ is convex. According to this property, we can use some efficient convex optimization algorithms to solve our objective function.

To handle the $\ell_1$ terms in function in Eq. (8),we introduce an auxiliary variable $\mathbf{U_X}$ as follows:

$$\begin{aligned} \min_{\mathbf{Z}\in\Omega_2,\mathbf{U_X}\geq\mathbf{0}} \; & J_2^{'}(\mathbf{Z},\mathbf{U_X}) = -\mathrm{tr}\mathbf{Y}_\mathbf{w}^\top\mathbf{Z} + \lambda_2\mathrm{tr}\left(\overline{\mathbf{G}}^\top\mathbf{U_X}\right) \\ & \text{s.t.}\mathbf{AZ} \leq \mathbf{U_X}, -\mathbf{AZ} \leq \mathbf{U_X}, \mathbf{Z}\cdot\mathbf{1}_m = \mathbf{1}_u \end{aligned} \tag{17}$$

Where $\mathbf{U_X} \in \mathbb{R}^{u_e\times m}$ with $\mathbf{U_X}(e,k) = |\mathbf{Z}_{e_1k} - \mathbf{Z}_{e_2k}|$. The edge $e$ is from $e_1$ to $e_2$ in the graph $\mathcal{G}$. $\overline{\mathbf{G}} = (\mathbf{g},\cdots,\mathbf{g}) \in \mathbb{R}^{u_e\times m}$, $\mathbf{g}(e) = \mathbf{G}(e_1,e_2), \forall e \in \mathcal{E}$. $\mathbf{A} \in \{-1,1,0\}^{u_e\times u}$ with $\mathbf{A}(e,e_1) = 1, \mathbf{A}(e,e_2) = -1$ for $e \in \mathcal{E}$ and $e_1, e_2 \in \mathcal{U}$, while other entries being 0.

LP problem in Eq. (17) can be solved by many off-the-shelf standard algorithms. However, most existing solvers can only be applied directly to vector variables. Although we can vectorize Eq. (17), this will cause a large LP problem, which is a standard but inefficient solution. Instead, we use alternating direction method of multipliers (ADMM) [28], which has good convergence and can take advantage of the special structural characteristics of our problem. In addition, in order to accelerate the convergence rate of the algorithm, we add a small enough number to the row equality constraint of the label matrix in problem (15) to make a slight change to the original problem. We satisfy the row of $\mathbf{Z} : (1+\epsilon)\mathbf{1}_u \leq \mathbf{Z}\cdot\mathbf{1}_m \leq (1-\epsilon)\mathbf{1}_u$. It is noted that this slight float will not have any impact when we restore the obtained soft label to the hard label, and will not cause a big error in the iteration process, the convergence speed of problem in (17) can be improved by

$$\begin{aligned} \min_{\mathbf{Z}\in\Omega_2,\mathbf{U_X}\geq\mathbf{0}} \; & J_2^{'}(\mathbf{Z},\mathbf{U_X}) \\ \text{s.t.}\mathbf{AZ} \leq \mathbf{U_X}, & -\mathbf{AZ} \leq \mathbf{U_X}, (1+\epsilon)\mathbf{1}_u \leq \mathbf{Z}\cdot\mathbf{1}_m \leq (1-\epsilon)\mathbf{1}_u \end{aligned} \tag{18}$$

According to the traditional ADMM method, we first introduce two relaxation variables $\Phi_1$, $\Phi_2$ and give the augmented Lagrangian function of the problem in Eq. (18).

$$\begin{aligned} &\mathcal{L}_{\rho_1,\rho_2}(\mathbf{Z},\mathbf{U_X},\Lambda_1,\Lambda_2,\Phi_1,\Phi_2) = -\mathbf{Y}_\mathbf{w}^\top\mathbf{Z} + \lambda_2\mathrm{tr}\left(\overline{\mathbf{G}}^\top\mathbf{U_X}\right) \\ &+ \mathrm{tr}[\Lambda_1^\top(\overline{\mathbf{A}}\mathbf{Z} - \overline{\mathbf{C}}\mathbf{U_X} - \overline{\mathbf{H}} + \Phi_1)] + \mathrm{tr}[\Lambda_2^\top(\mathbf{Z}\overline{\mathbf{B}} - \overline{\mathbf{D}} + \Phi_2)] \\ &+ \frac{\rho_1}{2}\|\overline{\mathbf{A}}\mathbf{Z} - \overline{\mathbf{C}}\mathbf{U_X} - \overline{\mathbf{H}} + \Phi_1\|_F^2 + \frac{\rho_2}{2}\|\mathbf{Z}\overline{\mathbf{B}} - \overline{\mathbf{D}} + \Phi_2\|_F^2 \end{aligned} \tag{19}$$

where $\overline{\mathbf{A}} = [\mathbf{A}; -\mathbf{A}; -\mathbf{1}_{1\times u}; \mathbf{1}_{1\times u}], \overline{\mathbf{C}} = [\mathbf{I}_{u_e}; \mathbf{I}_{u_e}; \mathbf{0}_{2\times n_e}]$, $\overline{\mathbf{H}} = [\mathbf{0}_{2n_e\times m}; -\boldsymbol{v}_1; \boldsymbol{v}_2], \overline{\mathbf{B}} = [-\mathbf{1}_m, \mathbf{1}_m], \overline{\mathbf{D}} = [-(1-\epsilon)\mathbf{1}_n, (1+\epsilon)\mathbf{1}_n], \Phi_1 \in \mathbb{R}^{2(u_e+1)\times m}, \Phi_2 \in \mathbb{R}^{u\times 2}$. And $\Lambda_1 \in \mathbb{R}^{2(u_e+1)\times m}, \Lambda_2 \in \mathbb{R}^{u\times 2}$ represent two Lagrangian parameter matrices, while $\rho_1, \rho_2 \geq 0$ denote the trade-off parameters for the two augmented terms. According to ADMM, each parameter in the problem (19) can be updated by the following steps iteratively.

$$\begin{aligned} \mathbf{Z}^{t+1} = \arg\min_{\mathbf{Z}\in[0,1]} \; & \mathrm{tr}(\mathbf{M}_0^\top\mathbf{Z}) + \frac{\rho_1}{2}\mathrm{tr}(\mathbf{Z}^\top\mathbf{M}_1\mathbf{Z}) \\ & + \rho_2\mathrm{tr}(\mathbf{Z}\cdot\mathbf{1}_{m\times m}\cdot\mathbf{Z}^\top) \end{aligned} \tag{20}$$

$$\mathbf{U}_\mathbf{X}^{t+1} = \max\{0, -\frac{1}{2\rho_1}\mathbf{M_X}\} \tag{21}$$

$$\Phi_1^{t+1} = \max\{0, \overline{\mathbf{C}}\mathbf{U}_\mathbf{X}^{t+1} + \overline{\mathbf{H}} - \overline{\mathbf{A}}\mathbf{Z}^{t+1} - \frac{1}{\rho_1}\Lambda_1^t\} \tag{22}$$

$$\Phi_2^{t+1} = \max\{0, \overline{\mathbf{D}} - \mathbf{Z}^{t+1}\overline{\mathbf{B}} - \frac{1}{\rho_2}\Lambda_2^t\} \tag{23}$$

$$\Lambda_1^{t+1} = \Lambda_1^t + \rho_1(\overline{\mathbf{A}}\mathbf{Z}^{t+1} - \overline{\mathbf{C}}\mathbf{U}_\mathbf{X}^{t+1} - \overline{\mathbf{H}} + \Phi_1^{t+1}) \tag{24}$$

$$\Lambda_2^{t+1} = \Lambda_2^t + \rho_2(\mathbf{Z}^{t+1}\overline{\mathbf{B}} - \overline{\mathbf{D}} + \Phi_2^{t+1}) \tag{25}$$

where $\mathbf{M}_0 = -\mathbf{Y}_\mathbf{w} + \overline{\mathbf{A}}^\top[\Lambda_1^t - \rho_1(\overline{\mathbf{C}}\mathbf{U}_\mathbf{X}^t + \overline{\mathbf{H}} - \Phi_1^t)] + [\Lambda_2^t - \rho_2(\overline{\mathbf{D}} - \Phi_2^t)]\overline{\mathbf{B}}^\top, \mathbf{M}_1 = \overline{\mathbf{A}}^\top\overline{\mathbf{A}}, \mathbf{M_X} = \lambda_2\overline{\mathbf{G}} - \overline{\mathbf{C}}^\top[\Lambda_1^t + \rho_1(\overline{\mathbf{A}}\mathbf{Z}^{t+1} - \overline{\mathbf{H}} + \Phi^t)]$.

**Algorithm 2** Learning method of the complete label matrix

**Input**: Initial $\{\mathbf{U}^0, \mathbf{Z}^0, \mathbf{U}_\mathbf{x}^0, \Phi^0, \Lambda_1^0, \Lambda_2^0\}$
**Output**: $\mathbf{Z}^*\}$
1: **while** not converged **do**
2: update $\mathbf{Z}^{t+1}$ by solving Eq. (20);
3: update $\mathbf{U}_\mathbf{X}^{t+1}$ by solving Eq. (21);
4: update $\Phi^{t+1}$ by solving Eq. (22);
5: update $\Lambda_1^{t+1}, \Lambda_2^{t+1}$ by solving Eq. (24) and Eq. (25).
8: **end while**
9: $\mathbf{Z}^* = \mathbf{Z}^{t+1}$

After the above update, we get the complete label matrix. Algorithm 2 briefly summarizes the above steps.

So far, we have completed the iterative solution of the whole model. In general, we complete label propagation the training of classifier by solving $\mathbf{Z}$ and $\mathbf{W}$ iteratively.

## 5 Discussions

In this section, we will analyze the computational complexity and convergence of the model, then discuss the influence of prior information accuracy and parameter determination.

### 5.1 Computational Complexity

In our algorithm, there are two computations that need to be discussed in particular. The first is the calculation of the gradient and the Hessian matrix in formula in Eq. (16), where we use the vectorization technique to avoid the multiplication of the high-dimensional matrix. We only need $\mathcal{O}((d+m)nk+(n+L)k^2)$ time to compute both $\nabla g(\boldsymbol{u})$ and $\nabla^2 g(\boldsymbol{u})\boldsymbol{s}$ efficiently. The second part is the large number of operations in ADMM solution. While we notice that in updating formulas in Eqs. (20)-(25), only the update of $\mathbf{Z}^{t+1}$ is effectively solved by the projected gradient descent algorithm (PGD) using the accurate linear search method, while other variables have closed solutions. Since ADMM only involves matrix multiplication, its computational complexity is about $\mathcal{O}\left(T\left(amu^2+bm^2u+cmu\right)\right)$, where $T$ is the number of iterative steps and and $a, b, c$ are small scalars.

### 5.2 Convergence Analysis

In the alternating iteration in section 4.1, it is easy to know from formula in Eq.(11) that the objective function declines in each iteration, i.e.

$$J_1(\mathbf{U}^t,\mathbf{V}^t) \leq J_1(\mathbf{U}^{t-1},\mathbf{V}^t) \leq J_1(\mathbf{U}^{t-1},\mathbf{V}^{t-1})$$

In addition, due to the object is a convex function, the sequence $\{(\mathbf{U}^t,\mathbf{V}^t)\}$ will converge to a stationary point when the minimum for both $J_1(\mathbf{U}^{t-1},\mathbf{V})$ and $J_1(\mathbf{U},\mathbf{V}^t)$ are uniquely defined [39]. According to [28], the solution of ADMM for convex problems in Section 4.2 can been proved to obtain the optimal. We give the following detailed convergence lemma:

**Lemma 3.** *Assumption that 1) the (extended-real-valued) functions $f(\boldsymbol{x}):\mathbb{R}^n \to \mathbb{R}\cup\{+\infty\}$ and $g(\boldsymbol{z}):\mathbb{R}^m \to \mathbb{R}\cup\{+\infty\}$ are closed, proper, and convex; 2) The augmented Lagrangian $\mathcal{L}$ has a saddle point, explicitly, there exist $(\boldsymbol{x}^\star,\boldsymbol{y}^\star,\boldsymbol{y}^\star)$, for which $\mathcal{L}(\boldsymbol{x}^\star,\boldsymbol{z}^\star,\boldsymbol{y}) \leq \mathcal{L}(\boldsymbol{x}^\star,\boldsymbol{z}^\star,\boldsymbol{y}^\star) \leq \mathcal{L}(\boldsymbol{x},\boldsymbol{z},\boldsymbol{y}^\star)$ holds for all $\boldsymbol{x},\boldsymbol{z},\boldsymbol{y}$ ($\boldsymbol{y}$ is the dual variable). Then the ADMM iterates satisfy the following: residual convergence, objective convergence and dual variable convergence.*

Specifically, we have the residual $r_1 = \|\overline{\mathbf{A}}\mathbf{Z} - \overline{\mathbf{C}}\mathbf{U}_{\mathbf{X}} - \overline{\mathbf{H}} + \Phi_1\|_F^2$ and $r_2 = \|\mathbf{Z}\overline{\mathbf{B}} - \overline{\mathbf{D}} + \Phi_2\|_F^2$ tends to 0 as the number of iteration steps increases, i.e., the iterates approach are feasible and shows that the row and column constraints of the label matrix are satisfied. Then the objective function $J_2^{'}(\mathbf{Z},\mathbf{U}_{\mathbf{X}})$ converges to the optimal value. In addition, we can know that for the dual problem, the dual variable $\Lambda_1$ and $\Lambda_2$ converges to a optimal solution.

### 5.3 Parameter Determination

We believe that the prior information is a interval centered on the accurate label ratio, and the precision means the length of the interval. Generally speaking, the higher the accuracy of prior information is, the uncertainty contained in the prior information will also have less impact on the model, so that the better performance of the algorithm will be. However, we know that the given prior information may not be accurate enough in precision, but it will not be serious deviation, so the influence of this information on the model is always positive, and intuitively, our algorithm is sensitive to the accuracy of prior information.

In the last, we discuss some important parameter selection problems in our model. Parameter selection is of great significance, which is still an open question up to now. There are three parameters: the trade-off parameters $\lambda_1$, $\lambda_2$ and the low rank constraint number $k$. For the first two parameters, we determine them by grid search in our work. For the latter, we will fix it experimentally, and just have to make it a small positive integer. We need to make sure every item of the objective function properly without seriously affecting the overall hypothesis, so the values of these two parameters are not very large.

## 6 Experiments

In this section, we thoroughly evaluate the effectiveness of our proposed algorithm with eight benchmark data sets, all of which are available at UCI machine learning repository[1]. First, we compare the performance of SSLLP with classical and some state-of-the-art semi-supervised classification algorithms. Then we use additional experiments to verify the three properties we discussed in section 5, i.e., algorithmic convergency, the influence of the prior information and parametric sensitivity.

### 6.1 Experimental Setting

Eight benchmark data sets used in our experiments are summarized as follows:

- The Vehicle data set contains 435 2D vehicle images in total which may be viewed from one of many different angles. The purpose is to classify a given silhouette as different types of vehicle, using a set of features extracted from the silhouette. The number ratio of maximum to minimum classes (NRC) is 1.05.
- The King-Rook vs. King-Pawn (KRvsKP) data set is a Chess End-Game: King+Rook versus King+Pawn on KRKPA7, which contains 3196 games. The pawn is the King + Rook's side (white) to move. The result is either that pawn side loses or wins. The NRC is 1.09.
- The Wine data set is the result of a chemical analysis of 130 wines grown in the same region in Italy but derived from different cultivars. The analysis determined the quantities of 13 constituents found in each of wines. The NRC is 1.20.
- The Liver Disorders data set is a medical data set containing 345 samples. It describes the effects of different test results, such as blood tests, on liver disorders. The NRC is 1.38.
- The Balance data set is generated to model psychological experimental results, containing 625 samples. Each example is classified as having the balance scale tip to the right, tip to the left. The attributes are the left weight, the left distance, the right weight, and the right distance. The NRC is 5.88.

1. https://archive.ics.uci.edu/ml/datasets.php

Table 2
Accuracies on the various data sets with 4%, 6%, 8%, 10% and 12% labeled instances on eight UCI data sets. The best performance on each data set is bolded. •/○ indicates SSLLP is significantly better/worse than compared methods (paired t-tests at 95% significance level). The win/tie/loss counts for SSLLP are summarized in the last rows.

| Data set | Label | SVM | S4VM | LSR | GGSSL | SSODM | SSLLP |
|---|---|---|---|---|---|---|---|
| Vehicle | 4% | .741 ± .031 • | .776 ± .056 • | .787 ± .045 • | .714 ± .040 • | .781 ± .050 • | **.898 ± .037** |
| | 6% | .773 ± .053 • | .823 ± .046 • | .787 ± .028 • | .778 ± .061 • | .832 ± .052 • | **.907 ± .034** |
| | 8% | .813 ± .044 • | .867 ± .032 • | .810 ± .014 • | .822 ± .046 • | .857 ± .062 • | **.917 ± .035** |
| | 10% | .841 ± .023 • | .874 ± .047 • | .824 ± .034 • | .873 ± .052 • | .874 ± .053 • | **.928 ± .035** |
| | 12% | .876 ± .029 • | .884 ± .033 • | .864 ± .041 • | .890 ± .040 • | .898 ± .021 • | **.926 ± .031** |
| KRvsKP | 4% | .887 ± .012 • | .874 ± .013 • | .851 ± .055 • | .847 ± .055 • | .883 ± .039 • | **.922 ± .025** |
| | 6% | .901 ± .019 • | .893 ± .018 • | .889 ± .034 • | .873 ± .043 • | .905 ± .020 • | **.942 ± .021** |
| | 8% | .912 ± .009 • | .900 ± .013 • | .903 ± .036 • | .886 ± .035 • | .925 ± .014 • | **.957 ± .015** |
| | 10% | .916 ± .005 • | .928 ± .012 • | .920 ± .037 • | .908 ± .022 • | .932 ± .009 • | **.968 ± .010** |
| | 12% | .926 ± .007 • | .920 ± .010 • | .918 ± .027 • | .913 ± .016 • | .936 ± .007 • | **.970 ± .011** |
| Wine | 4% | .870 ± .061 • | .903 ± .084 • | .884 ± .059 • | .877 ± .065 • | .881 ± .044 • | **.973 ± .022** |
| | 6% | .855 ± .047 • | .922 ± .052 • | .901 ± .066 • | .906 ± .070 • | .891 ± .065 • | **.985 ± .021** |
| | 8% | .894 ± .079 • | .916 ± .051 • | .920 ± .053 • | .915 ± .052 • | .916 ± .051 • | **.979 ± .025** |
| | 10% | .909 ± .054 • | .929 ± .049 • | .936 ± .044 • | .917 ± .037 • | .929 ± .037 • | **.982 ± .026** |
| | 12% | .915 ± .056 • | .941 ± .050 • | .939 ± .066 • | .930 ± .044 • | .938 ± .031 • | **.985 ± .016** |
| Liver Disorders | 4% | .545 ± .071 • | .574 ± .059 • | .559 ± .065 • | .567 ± .077 • | .569 ± .060 • | **.628 ± .045** |
| | 6% | .559 ± .066 • | .609 ± .058 • | .586 ± .070 • | .571 ± .050 • | .602 ± .055 • | **.640 ± .043** |
| | 8% | .579 ± .073 • | .618 ± .051 • | .586 ± .052 • | .613 ± .057 • | .590 ± .059 • | **.663 ± .055** |
| | 10% | .597 ± .047 • | .614 ± .035 • | .588 ± .055 • | .621 ± .059 • | .623 ± .051 • | **.674 ± .049** |
| | 12% | .606 ± .058 • | .622 ± .035 • | .617 ± .049 • | .628 ± .049 • | .627 ± .050 • | **.687 ± .053** |
| WDBC | 4% | .856 ± .052 • | .886 ± .037 • | .878 ± .063 • | .889 ± .045 • | .887 ± .048 • | **.922 ± .034** |
| | 6% | .883 ± .039 • | .905 ± .042 • | .885 ± .033 • | .889 ± .037 • | .901 ± .031 • | **.935 ± .021** |
| | 8% | .872 ± .051 • | .911 ± .025 • | .898 ± .031 • | .910 ± .025 • | .907 ± .033 • | **.938 ± .028** |
| | 10% | .902 ± .028 • | .909 ± .048 • | .903 ± .021 • | .899 ± .016 • | .904 ± .029 • | **.940 ± .025** |
| | 12% | .905 ± .033 • | .916 ± .022 • | .911 ± .028 • | .907 ± .017 • | .906 ± .023 • | **.944 ± .019** |
| WPBC | 4% | .662 ± .050 • | .685 ± .063 • | .693 ± .058 • | .695 ± .052 • | .700 ± .046 • | **.753 ± .042** |
| | 6% | .714 ± .051 • | .709 ± .070 • | .714 ± .064 • | .711 ± .054 • | .717 ± .042 • | **.777 ± .056** |
| | 8% | .728 ± .050 • | .721 ± .044 • | .722 ± .051 • | .724 ± .053 • | .730 ± .047 • | **.778 ± .054** |
| | 10% | .718 ± .057 • | .737 ± .045 • | .738 ± .047 • | .734 ± .044 • | .735 ± .051 • | **.776 ± .055** |
| | 12% | .728 ± .034 • | .746 ± .043 • | .750 ± .040 • | .747 ± .039 • | .752 ± .031 • | **.780 ± .040** |
| Balance | 4% | .830 ± .069 • | .833 ± .066 • | .836 ± .024 • | .858 ± .028 | .845 ± .054 • | **.862 ± .033** |
| | 6% | .852 ± .030 • | .857 ± .041 • | .833 ± .023 • | .863 ± .036 • | .848 ± .042 • | **.877 ± .036** |
| | 8% | .857 ± .046 • | .849 ± .031 • | .852 ± .034 • | .861 ± .026 • | .885 ± .031 | **.886 ± .030** |
| | 10% | .856 ± .039 • | .868 ± .049 • | .868 ± .027 • | .867 ± .028 • | **.898 ± .042** | .893 ± .026 |
| | 12% | .861 ± .042 • | .882 ± .029 • | .866 ± .034 • | .875 ± .032 • | **.905 ± .041** | .899 ± .031 |
| Abalone | 4% | .906 ± .016 • | .916 ± .023 • | .920 ± .021 • | .918 ± .018 • | .916 ± .015 • | **.947 ± .017** |
| | 6% | .911 ± .015 • | .914 ± .019 • | .923 ± .021 • | .911 ± .013 • | .927 ± .012 • | **.945 ± .022** |
| | 8% | .910 ± .019 • | .925 ± .019 • | .923 ± .025 • | .922 ± .016 • | .932 ± .014 • | **.952 ± .019** |
| | 10% | .918 ± .025 • | .924 ± .019 • | .927 ± .015 • | .924 ± .023 • | .937 ± .020 • | **.962 ± .015** |
| | 12% | .913 ± .022 • | .929 ± .018 • | .930 ± .019 • | .938 ± .012 • | .938 ± .025 • | **.966 ± .016** |
| Transfusion | 4% | .696 ± .036 • | .668 ± .033 • | .727 ± .033 • | .718 ± .028 • | .687 ± .042 • | **.749 ± .029** |
| | 6% | .696 ± .032 • | .679 ± .039 • | .733 ± .035 • | .725 ± .033 • | .711 ± .035 • | **.752 ± .022** |
| | 8% | .718 ± .034 • | .695 ± .037 • | .755 ± .038 • | .742 ± .036 • | .710 ± .036 • | **.768 ± .020** |
| | 10% | .734 ± .033 • | .733 ± .029 • | .765 ± .032 • | .754 ± .031 • | .734 ± .030 • | **.775 ± .015** |
| | 12% | .755 ± .028 • | .753 ± .030 • | .774 ± .029 | .758 ± .026 • | .733 ± .035 • | **.774 ± .022** |
| Tic-Tac-Toe | 4% | .571 ± .028 • | .608 ± .036 • | .629 ± .020 • | .627 ± .022 • | .587 ± .026 • | **.649 ± .021** |
| | 6% | .591 ± .024 • | .617 ± .028 • | .627 ± .022 • | .609 ± .028 • | .589 ± .024 • | **.644 ± .021** |
| | 8% | .605 ± .025 • | .632 ± .033 • | .625 ± .035 • | .623 ± .032 • | .605 ± .024 • | **.648 ± .025** |
| | 10% | .614 ± .022 • | .641 ± .028 • | .633 ± .034 • | .642 ± .027 • | .610 ± .023 • | **.656 ± .020** |
| | 12% | .623 ± .030 • | .648 ± .023 • | .650 ± .027 • | .648 ± .030 • | .637 ± .025 • | **.667 ± .026** |
| German | 4% | .677 ± .036 • | .670 ± .029 • | .673 ± .033 • | .692 ± .026 | .677 ± .029 • | **.696 ± .033** |
| | 6% | .686 ± .019 • | .680 ± .016 • | .674 ± .019 • | .695 ± .034 • | .696 ± .031 • | **.704 ± .028** |
| | 8% | .705 ± .035 | .680 ± .025 • | .688 ± .026 • | .699 ± .025 | **.715 ± .029** | .712 ± .032 |
| | 10% | .694 ± .024 • | .687 ± .036 • | .696 ± .030 • | .703 ± .026 • | .710 ± .033 • | **.724 ± .025** |
| | 12% | .713 ± .028 • | .709 ± .030 • | .697 ± .026 • | .706 ± .017 • | .719 ± .024 • | **.728 ± .031** |
| SSLLP: w/t/l | | 54 / 1 / 0 | 55 / 0 / 0 | 54 / 1 / 0 | 52 / 3 / 0 | 51 / 4 / 0 | |

- The Abalone data set comes from an original (non-machine-learning) study: predicting the age of abalone from physical measurements. Furthermore, examples with missing values were removed and the ranges of the continuous values have been scaled. The original data set contains 4177 samples, and we selected the two categories with the most and least categories. The NRC is $10.28$.
- The Wisconsin diagnostic breast cancer (WDBC) dataset consists of the digitized image of a fine needle aspirate (FNA) of a breast mass from 357 patients with benign tumors and 212 patients with malignant tumors. The features are computed from the image that describes characteristics of the cell nuclei present, including radius, perimeter, compactness and so on.
- The Wisconsin prognostic breast cancer (WPBC) dataset were used to diagnose patients with recurrence within 24 months based on 30 characteristics, including 47 patients with recurrence and 151 patients without recurrence.
- The Transfusion data set is taken from the Blood Transfusion Service Center in Hsin-Chu City to demonstrate the RFMTC marketing model, 748 donors were randomly selected from the donor database. Each sample includes 4 characteristics, such as blood donation or not and total number of donation. The NRC is $3.20$.
- The Tic-Tac-Toe data set comes from a classification task on possible configurations of tic-tac-toe game, with a total of 958 games. This database encodes the complete set of possible board configurations at the end of tic-tac-toe games. The number of attributes are 9 and the NRC is $1.89$.
- The German data set classifies people described by a set of attributes as good or bad credit risks. There are a total of 1000 people, each of whom is represented by 24 attributes. The NRC is $2.33$.

For each data set, we randomly split it into two parts, with 75% of the samples for training and the rest for testing. In addition, we examine the performance of each approach with varying amount of labeled data, i.e., 4%, 6%, 8%, 10% and 12% of the training data as labeled data set. The remaining instances are regarded as the testing set.

We compare our method with 1) the baseline algorithm, standard SVM (using only the labeled training samples), and four state-of-the-art SSL algorithms, i.e., 2) safe SVM (S4VM); 3) semi-supervised classification via Local Spline Regression (LSR); 4) general graph-based semi-supervised learning (GGSSL) 5) Semi-Supervised Optimal Margin Distribution Machines (SSODM). Note that GGSSL is not an inductive algorithm as it is a graph-based label propagation approach, we use the predicted labels by the GGSSL on the transduction set to train a linear supervised SVM classifier (implemented with LIBLINEAR [40]) on the original data. Since only a part of unlabeled samples can be seen in the training phase, and the real purpose of learning is to classify the invisible samples, we mainly compared our method with the inductive learning algorithm in the experiment.

For all methods, we choose the best parameter settings by grid search strategy [18]. For our method, $\lambda_1$ and $\lambda_2$ need to determine the optimal values. We set their ranges from the set $\{10^{-2}, 10^{-1}, 1, 10^{1}, 10^{2}\}$. Meanwhile, the rank and the error of label matrix constraints are fixed as $k = \min\{10, m\}$ and $\epsilon = 10^{-2}$, respectively. We adopt two metrics the classification accuracy (ACC) and $F_{score}$ to evaluate the performance of all methods. For each data set, we calculate the average of $F_{score}$ for each category, denoted as $F_{mean}$. All experiments are repeated 20 times. We present the average values and their corresponding standard deviations of all methods on the test set.

### 6.2 Results

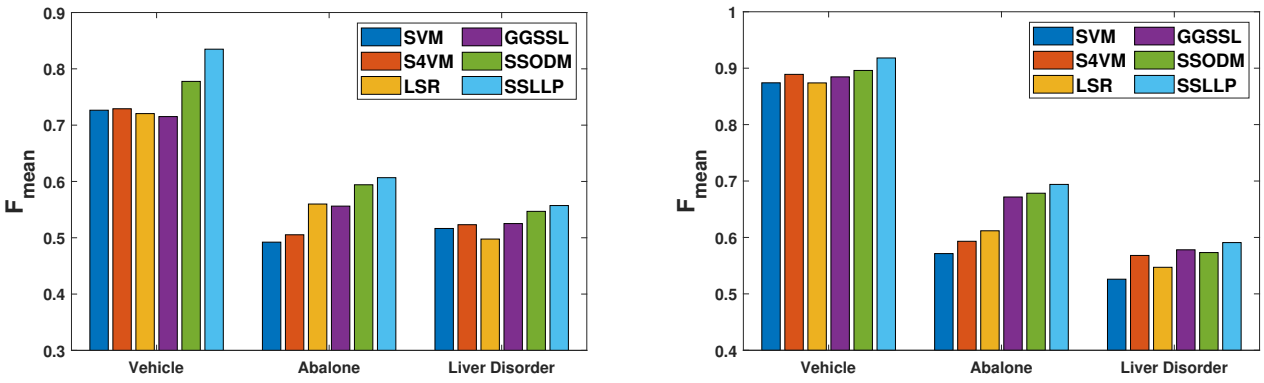
When comparing with other methods, we set the label proportion constraint to a 10% difference from the real label proportion. Table 2 shows the accuracies of SSLLP comparison to the supervised SVM and the four semi-supervised algorithms under five different markup ratios, the best performance on each data set is bolded. In addition, Figure 2 presents the results of $F_{mean}$ for all the comparison methods on three datasets. Results indicate that, compared with other semi-supervised learning approaches, SSLLP significantly improves the performance at almost all data sets under various markup ratios. Our method achieved the best results in seven of the eight data sets, and was similar to the best results in the rest data set Balance. This is because, on the one hand, we make effective use of the prior information of label proportionality; on the other hand, we consider many characteristics of semi-supervised learning itself, such as constraint of matrix rows and label discretization. Therefore, although our framework is a simple linear model, it still performs well. For all methods, it is obvious that the higher the markup ratio, the higher the classification accuracy. One interesting fact is that on most data sets, the advantages of our method are more obvious when the markup ratio is lower. This can be explained from the perspective of information theory. When the markup ratio is lower, the importance of label proportionality is higher. while When the markup ratio is higher, there will be some redundancy between the information carried by the data and the prior information, so the advantage of our method is not so great.

In addition, since SSODM considers the optimal margin distribution of SVM algorithm, it also achieves good results. The supervised SVM algorithm is generally unsatisfactory due to the relatively small sample size available, while semi-supervised learning can utilize the unlabeled data for model training to bring extra advantages.

### 6.3 Convergence

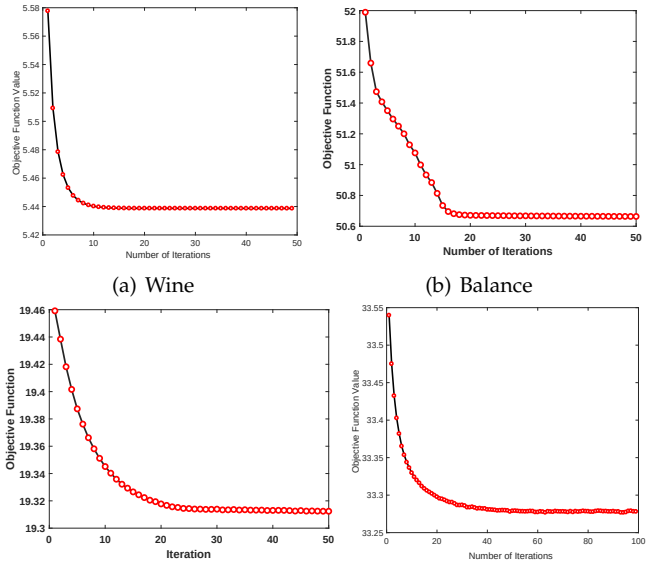
In Section 5, we theoretically analyzed that both the updating process of the classifier $\mathbf{W}$ and the label matrix $\mathbf{Z}$ converges. In this section, we experimentally verify the overall convergence of the algorithm. Figure 3 shows the convergence on the Wine and WDBC data sets, respectively. We find that it would tend to be stable after 100 iterations. In fact, for small sample data sets, such as Wine, we can ensure that it converges around 10 iterations.

Besides, in the process of updating $\mathbf{W}$ and $\mathbf{Z}$ respectively, we find that $\mathbf{W}$ converges rapidly. In fact, with the number of iterations of the overall cycle increasing, the convergence rate will be faster. For the solution of $\mathbf{Z}$, since our idea is updated




(a) 4% labeled instances (b) 12% labeled instances

Figure 2. The classification results measured by $F_{mean}$ of all compared methods on the three data sets (Vehicle, Abalone, Liver Disorder) under two different ratios of labeled instances: (a) 4% and (b) 12%.




(a) Wine (b) Balance

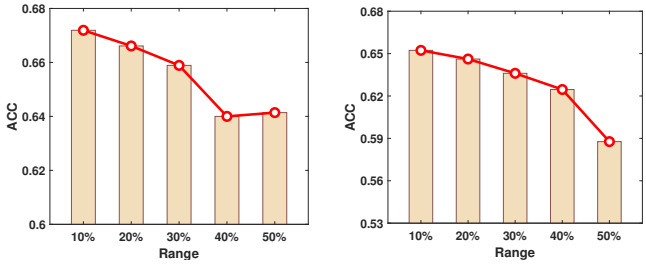
(c) Abalone (d) Vehicle

Figure 3. Number of iterations vs. the objective value of our proposed methods with 10% labeled data on four datasets: Wine and WDBC.

one by one according to the elements, the operation of high-dimensional matrix will be involved, so the convergence rate will be much slower. In fact, because of this reason, the speed of our algorithm will be affected to some extent. In the experiment, we also found that the update of relaxation variables $\Phi_1$, $\Phi_2$ are more and more close to 0, and the label matrix also gradually satisfies the row and column constraint.

### 6.4 Prior Information Impact Analysis

In this experiment, we verify the influence of prior information of different ranges on the performance of the model. We select five different ranges, that is, the upper and lower bounds of the constraint differ from the real label by 10%, 20%, 30%, 40% and 50%, respectively. The results are shown in Figure 4. An obvious fact is that the greater the difference between the upper and lower bounds of the constraint, the lower performance of the algorithm. This is consistent with our intuitive understanding, because the larger the range of constraints, the greater uncertainty it contains and the greater the error generated in the solving process. Furthermore, We noticed that when the constraint range increased from 10% to 50%, the performance of our algorithm decreased by nearly 5%, which indicated that our algorithm was sensitive to the constraint range. This also shows that our algorithm makes effective use of prior information. When the accuracy of prior information is improved, our algorithm will be improved certainly.



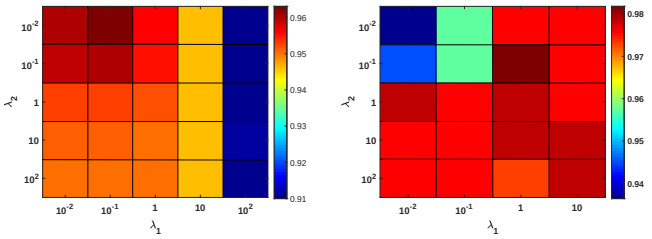

(a) Liver Disorder (b) Tic-Tac-Toe

Figure 4. The classification accuracy (ACC) results of our SSLLP with the change of label proportion of each class on two datasets: Liver Disorder and WPBC.

### 6.5 Parameter Sensitivity Analysis

Figure 5 shows the classification accuracy of SSLLP with varying value of the parameter $\lambda_1$ and $\lambda_2$ on two data sets. Other parameter settings are consistent with those in Section 6.2. One thing to note is that for the Wine data set, when $\lambda_1 = 10^2$, the accuracy drops so sharply that it is less than 0.5, so we just leave it out and report the rest of the results. It can be seen that when the values of $\lambda_1$ and $\lambda_2$ are set relatively small, our algorithm achieves satisfactory results, and the algorithm is not very sensitive to these two parameters. But when they're large, especially $\lambda_1$, the performance of our algorithm is significantly affected. This is because when $\lambda_1$ is very large, the hypothesis of label consistency will be seriously affected in the solution process, leading to a big difference between classification results and the reality. When $\lambda_1$ is very small, the performance of the algorithm is not affected seriously, which may be because the feature dimensions of the data sets we selected are not high, so the advantages brought by low-rank constraint are not obvious. However, when $\lambda_2$ varies, the effect on the algorithm is relatively small, because in the process of ADMM solving,

the Lagrangian parameter matrices $\Lambda_1$ will eliminate the impact brought by $\lambda_2$ to some extent.




(a) Abalone (b) Wine

Figure 5. Effect of parameters $\lambda_1$ and $\lambda_2$ with respect to ACC on two datasets. Different colors mean different ACC values, and the color close to red indicates high scores.

### 6.6 Computational Time Comparison

In this subsection, we provide some experimental results to compare the computational time. To show the influence of data size $n$ and dimensionality $d$, we select two data sets, i.e., Coil20 and CovType, since they have the largest n and d among eight data sets. With a naive MATLAB implementation, we report the computational time of ISR and other feature selection methods. We have tested all the algorithms on a laptop with 8 processors (2.27 GHz for each) and 32 GB RAM memory. The results are shown in Table 4. The experimental results of Table 4 indicate that costs the least time in all the cases, InfFS and S2FS2R costs the most time in most of cases. The computational costs of our ISR change with different number of instances and different dimension of features.

Table 3
Computational time (CPU time in second) of different methods on Abalone, KRvsKP and WPBC.

| Data | S4VM | LSR | GGSSL | SSODM | SSLLP | |
|---|---|---|---|---|---|---|
| WPBC | 2.02 | 0.16 | 0.02 | 0.01 | 2.33 | |
| KRvsKP | 262.68 | 4.44 | 0.13 | 0.10 | 326.60 | |
| Abalone | 290.75 | 4.46 | 0.30 | 0.15 | 382.41 | |

## 7 Conclusions

In this paper, we study the learning problem with the prior label proportion information, and then propose a novel SSL method named as SSLLP. By imposing the cardinality bound constraints, our method can preserve label consistency and label proportion information to enhance its classification performance. Based on Lovász extension of submodular function, our original mixed-integer constrained minimization problem can be transformed into a convex optimization problem. And then an efficient optimization algorithm is proposed to solve it. Finally, empirical studies present our SSLLP can significantly improve the classification performance by effectively utilizing prior information. In this paper, we only focus on the classification problem with label proportion. In the future, how to solve other tasks, such as clustering and ranking etc. is an interesting future work. Besides, how to design more efficient optimization algorithm is also worth studying.

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